POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define Microcontroller. What are the essential difference between Von Neumann and Harvard architecture? 7
   b) Explain the architecture of 8085 microprocessor? 8

2. a) Draw the timing diagrams for the instruction STA E040H. The address and opcode of the instruction is shown below: 8

Address	Mnemonics	Op code
30FF	STA E040H	32H
3100		40H
3101		41H

   b) What are the types of instruction depending upon word size? Explain different type of addressing modes of 8085 microprocessor. 7

3. a) Draw and explain the functional block diagram of 8259 - PIC. 8
   b) What do you mean by unique and non-unique address decoding? Differentiate between synchronous and asynchronous bus. 7

4. a) Write an 8086 ALP to check whether a given string is palindrome or not. 7
   b) What do you understand by memory interfacing? Design an interfacing circuit to interface one 4KB EPROM and two 2KB RAM for 8085 microprocessor using 3*8 decoder. 8

5. a) What is interrupt vector table? Explain procedures and macros. 7
   b) Compare RS232 and RS422. Explain how two computers can communicate with each other using RS232 standard. 8

6. a) What is the Asynchronous serial data communication? Compare serial and parallel communication. Draw the Block diagram of Intel 8251A USART and explain functionally in detail. 8
   b) Write the RTL for the instruction LXI D, 9050H in 8085 microprocessor. Explain the operations and uses of DAA, SHLD and LHLD instructions in 8085. 7

7. Write short notes on: (Any two) 2×5
   a) Memory mapped I/O and I/O mapped I/O
   b) Handshaking signals
   c) Types of interrupt.

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